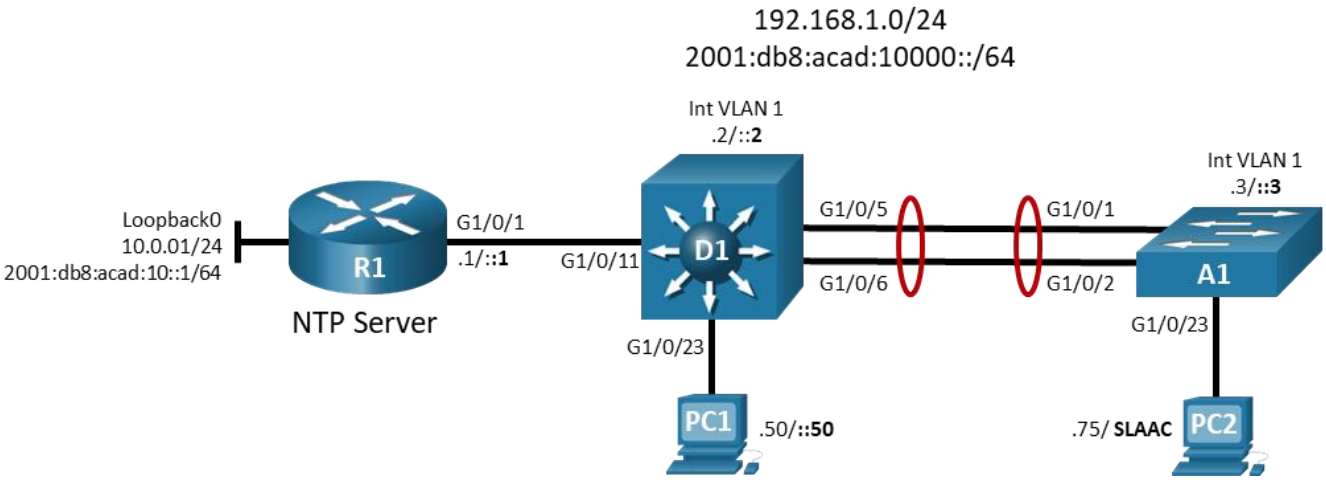
Lab - Implement Flexible Netflow

Topology



Addressing Table

Device	Interface	IP Address	IPv6 Address	IPv6 Link Local
R1	G0/0/1	192.168.1.1/24	2001:db8:acad:1000::1/64	fe80::1:1
	Loopback0	10.0.0.1/24	2001:db8:acad:10::1/64	fe80::1:2
D1	VLAN 1	192.168.1.2/24	2001:db8:acad:1000::2/64	fe80::d1:1
A1	VLAN 1	192.168.1.3/24	2001:db8:acad:1000::3/64	fe80::a1:1
PC1	NIC	192.168.1.50/24	2001:db8:acad:1000::50/64	EUI-64
PC2	NIC	192.168.1.75/24	Assigned by SLAAC	EUI-64

Objectives

Part 1: Build the Network and Configure Basic Device Settings and Interface Addressing

Part 2: Configure and Verify Flexible Netflow

Background / Scenario

NetFlow is a Cisco IOS feature that summarizes packet activity into flows so you can understand who is talking to whom, on which ports, and for how long. Instead of inspecting every packet, the router or switch groups packets with the same key attributes into a single flow and records statistics for that flow. This creates rich operational data while keeping overhead manageable.

A flow is typically identified by a 5-tuple: source IP, destination IP, source port, destination port, and IP protocol. Devices maintain a flow cache, update counters as packets pass, and export flow records at intervals to an external collector for storage and analysis. The resulting data supports network and security

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monitoring, capacity planning, traffic analysis, and IP accounting. Common tasks include identifying top talkers, spotting unusual traffic spikes, and understanding application usage across interfaces.

Flexible NetFlow extends the original NetFlow by letting you choose exactly which fields to match on and which statistics to collect. It uses a template-driven approach so you can tailor records to the applications and policies that matter in your environment. This flexibility makes it easier to align visibility with compliance needs, security workflows, and troubleshooting goals.

In this lab, you will configure and verify Flexible NetFlow using a flow record, monitor, and exporter to collect and export traffic data. You will then validate results by reviewing the flow cache and exported records.

Note: This lab is an exercise in configuring options available for Flexible Netflow and does not necessarily reflect network troubleshooting best practices.

Note: The routers used with CCNP hands-on labs are Cisco 8200L with Cisco IOS XE Release 17.12.05c (universalk9 image). The switches used in the labs are Cisco Catalyst 9300L with Cisco IOS XE Release 17.12.05 (universalk9 image) and Cisco Catalyst 9200L with Cisco IOS XE Release 17.12.05 (universalk9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs.

Note: IOS XE does not support classic Netflow. If your lab has ISR G2 series routers, skip Part 2 of this lab and do Part 3, which covers classic Netflow.

Note: Make sure that the switches have been erased and have no startup configurations. If you are unsure, contact your instructor.

Note: The default Switch Database Manager (SDM) template on a switch, such as Catalyst 2960, does not support IPv6. You must change the default SDM template to the dual-ipv4-and-ipv6 default template using the **sdm prefer dual-ipv4-and-ipv6 default** global configuration command. Changing the template will require a reboot.

Required Resources

- 1 Router (Cisco 8200L with Cisco IOS XE Release 17.12.05c universal image or comparable)
- 1 Switch (Cisco 9300L with Cisco IOS XE Release 17.12.05 universal image or comparable)
- 1 Switch (Cisco 9200L with Cisco IOS XE Release 17.12.05 universal image or comparable)
- 2 PCs (Choice of operating system with terminal emulation program and a packet capture utility installed, such as Wireshark)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Instructions

Part 1: Build the Network and Configure Basic Device Settings and Interface Addressing

In Part 1, you will set up the network topology and configure basic settings and interface addressing on routers.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram and cable as necessary.

Step 2: Configure basic settings for each device.

a. Console into each device, enter global configuration mode, and apply the basic settings. The startup configurations for each device are provided below.

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Router R1

```
hostname R1
no ip domain lookup
ipv6 unicast-routing
username admin privilege 15 secret cisco123
banner motd # R1, Implement Flexible Netflow #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 exec-timeout 0 0
 login local
 exit
interface g0/0/1
 ip address 192.168.1.1 255.255.255.0
 ipv6 address fe80::1:1 link-local
 ipv6 address 2001:db8:acad:1000::1/64
 no shutdown
 exit
interface loopback 0
 ip address 10.0.0.1 255.255.255.0
 ipv6 address fe80::1:2 link-local
 ipv6 address 2001:db8:acadd:10::1/64
 no shutdown
 exit
ntp master 3
ip domain name CCNP
crypto key generate rsa modulus 2048
end
```

Switch D1

```
hostname D1
no ip domain lookup
ipv6 unicast-routing
banner motd # D1, Implement Flexible Netflow #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 exec-timeout 0 0
 password cisco123
```

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```
 login
 exit
interface vlan 1
 ip address 192.168.1.2 255.255.255.0
 ipv6 address fe80::d1:1 link-local
 ipv6 address 2001:db8:acad:1000::2/64
 no shutdown
 exit
ip default-gateway 192.168.1.1
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
 exit
interface g1/0/23
 spanning-tree portfast
 switchport mode access
 no shutdown
 exit
interface g1/0/11
 spanning-tree portfast
 switchport mode access
 no shutdown
 exit
interface range g1/0/5-6
 switchport mode trunk
 channel-group 1 mode active
 no shutdown
 exit
ntp server 192.168.1.1
end
```

Switch A1

```
hostname A1
no ip domain lookup
ipv6 unicast-routing
banner motd # A1, Implement Flexible Netflow #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 exec-timeout 0 0
 password cisco123
 login
 exit
interface vlan 1
```

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```
 ip address 192.168.1.3 255.255.255.0
 ipv6 address fe80::a1:1 link-local
 ipv6 address 2001:db8:acad:1000::3/64
 no shutdown
 exit
ip default-gateway 192.168.1.1
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
 exit
interface range g1/0/1-2
 switchport mode trunk
 channel-group 1 mode active
 no shutdown
 exit
interface g1/0/23
 switchport mode access
 spanning-tree portfast
 no shutdown
 exit
ntp server 192.168.1.1
end
```

b. Set the clock on each device to UTC time.

c. Save the running configuration to startup-config.

d. Configure IPv4 and IPv6 addresses on hosts PC1 and PC2 as shown in the addressing table.

e. Verify that R1, D1, A1, and PC2 can successfully ping PC1 at 192.168.1.50.

Part 2: Configure and Verify Flexible NetFlow

As previously stated, Flexible NetFlow provides the ability to customize traffic analysis parameters. The workflow for Flexible NetFlow consists of four steps:

Step 1. Create Flow Records. Flow records define the information to be collected. There are predefined flow records so you can configure your own custom flow record to suit your needs.

Step 2. Create Flow Exporter. This defines where compiled statistical information is sent.

Step 3. Create Flow Monitor and associate Flow Records and Flow Exporters with it.

Step 4. Configure the appropriate interface for input or output caching associated with the appropriate Flow Monitor.

In this part of the lab, you will configure Flexible NetFlow to send statistical information about R1 interface g0/0/1 to PC1.

Step 1: Create flow records.

a. For our first flow record, we will use the predefined **ipv4 original-input** flow record. Because it is predefined, there is no configuration necessary.

b. For our second flow record, we will create a custom flow record. Because the first flow record is focused on input traffic, the second will focus on output traffic. Create a flow record named CCNP-CUSTOM-OUT.

```
R1(config)# flow record CCNP-CUSTOM-OUT
```

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1) Give the flow record a description.

```
R1(config-flow-record)# description Custom Flow Record for outbound traffic
```

2) Set up the flow record to match ipv4 destination address and transport destination.

```
R1(config-flow-record)# match ipv4 destination address
R1(config-flow-record)# match transport destination-port
```

3) Set up the flow record to collect bytes and packets.

```
R1(config-flow-record)# collect counter bytes
R1(config-flow-record)# collect counter packets
```

4) Use the **show flow record CCNP-CUSTOM-OUT** command to examine the results.

```
R1# show flow record CCNP-CUSTOM-OUT
flow record CCNP-CUSTOM-OUT:
  Description:        Custom Flow Record for outbound traffic
  No. of users:       0
  Total field space:  14 bytes
  Fields:
    match ipv4 destination address
    match transport destination-port
    collect counter bytes
    collect counter packets
```

Step 2: Create a flow exporter.

a. The flow exporter configuration defines where the cached information will be sent. Create a flow exporter named CCNP-COLLECTOR-HOST. Further specify that the exporter should use Netflow version 9, and point to 192.168.1.50 udp port 9999.

```
R1(config)# flow exporter CCNP-COLLECTOR-HOST
R1(config-flow-exporter)# destination 192.168.1.50
R1(config-flow-exporter)# export-protocol netflow-v9
R1(config-flow-exporter)# transport UDP 9999
R1(config-flow-exporter)# exit
```

b. Use the **show flow exporter CCNP-COLLECTOR-HOST** command to examine the results.

```
R1# show flow exporter CCNP-COLLECTOR-HOST
Flow Exporter CCNP-COLLECTOR-HOST:
  Description:              User defined
  Export protocol:          NetFlow Version 9
  Transport Configuration:
    Destination IP address: 192.168.1.50
    Source IP address:      192.168.1.1
    Transport Protocol:     UDP
    Destination Port:       9999
    Source Port:            63275
    DSCP:                   0x0
    TTL:                    255
    Output Features:        Used
```

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Step 3: Create flow monitors.

The flow monitor associates a flow record with the flow exporter. For our exercise, we need to create two flow monitors, one for each flow record.

a. Create the first flow monitor and name it CCNP-INBOUND-MONITOR using the **flow monitor CCNP-INBOUND-MONITOR** command. As part of the flow monitor, specify that it will record the netflow ipv4 original-input flow record, export the cache to the exporter every 30 seconds, and identify CCNP-COLLECTOR-HOST as the exporter.

```
R1(config)# flow monitor CCNP-INBOUND-MONITOR
R1(config-flow-monitor)# record netflow ipv4 original-input
R1(config-flow-monitor)# cache timeout active 30
R1(config-flow-monitor)# exporter CCNP-COLLECTOR-HOST
R1(config-flow-monitor)# exit
```

b. Create the second flow monitor and name it CCNP-OUTBOUND-MONITOR using the **flow monitor CCNP-OUTBOUND-MONITOR** command. As part of the flow monitor, specify that it will record the CCNP-CUSTOM-OUT flow record, export the cache to the exporter every 30 seconds, and identify CCNP-COLLECTOR-HOST as the exporter.

```
R1(config)# flow monitor CCNP-OUTBOUND-MONITOR
R1(config-flow-monitor)# record CCNP-CUSTOM-OUT
R1(config-flow-monitor)# cache timeout active 30
R1(config-flow-monitor)# exporter CCNP-COLLECTOR-HOST
R1(config-flow-monitor)# exit
```

c. Use the **show flow monitor** command to examine the results.

```
R1# show flow monitor
Flow Monitor CCNP-INBOUND-MONITOR:
  Description:       User defined
  Flow Record:       netflow ipv4 original-input
  Flow Exporter:     CCNP-COLLECTOR-HOST
  Cache:
    Type:                normal (Platform cache)
    Status:              not allocated
    Size:                200000 entries
    Inactive Timeout:    15 secs
    Active Timeout:      30 secs
    Trans end aging:   off

Flow Monitor CCNP-OUTBOUND-MONITOR:
  Description:       User defined
  Flow Record:       CCNP-CUSTOM-OUT
  Flow Exporter:     CCNP-COLLECTOR-HOST
  Cache:
    Type:                normal (Platform cache)
    Status:              not allocated
    Size:                200000 entries
    Inactive Timeout:    15 secs
    Active Timeout:      30 secs
    Trans end aging:   off
```

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Step 4: Configure the interface for flow caching.

The last step is to configure the appropriate interface(s) so that they will cache information. In our lab, we will focus on the input and output from interface g0/0/0 on R1. Use the **ip flow monitor <name> <direction>** command on g0/0/1 to specify the inbound and outbound flow monitors you have created.

```
R1(config)# interface g0/0/1
R1(config-if)# ip flow monitor CCNP-INBOUND-MONITOR input
R1(config-if)# ip flow monitor CCNP-OUTBOUND-MONITOR output
R1(config-if)# exit
```

Step 5: Create some traffic.

To gather statistics, we will need some traffic.

a. From PC2, open two Command Prompt windows and start a continuous ping to R1 using IPv4 and IPv6. As a part of each set of ping parameters, set the size of the packets to 1475 bytes.

   The Windows commands are as follows:

```
C:\> ping 10.0.0.1 -t -l 1475
C:\> ping 2001:db8:acad:1000::1 -t -l 1475
```

b. From switch A1, SSH to R1 using the user account **admin** / **cisco123**. Login and leave the session running.

c. From switch D1, use the extended ping utility to send pings to R1 Loopback 0 using a sweep range of 36 bytes to 18024 bytes. Set the repeat count to 1,000,000 and the sweep interval to 1.

```
D1# ping
Protocol [ip]:
Target IP address: 10.0.0.1
Repeat count [5]: 100000
Datagram size [100]:
Timeout in seconds [2]:
Extended commands [n]: y
Ingress ping [n]:
Source address or interface:
DSCP Value [0]:
Type of service [0]:
Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0x0000ABCD]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]: y
Sweep min size [36]:
Sweep max size [18024]:
Sweep interval [1]:
Type escape sequence to abort.
Sending 89945, [36..18024]-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
```

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d. On PC1, start Wireshark and apply the filter **ip.src == 192.168.1.1 && udp.dstport == 9999 && ! icmp**. This filter displays packets with a source IP of 192.168.1.1 that are destined for UDP port 9999, and it excludes ICMP traffic. Recall that port 9999 is what you configured for transporting NetFlow statistics.

Step 6: Wait 60 seconds then examine the results.

a. On PC1, observe the Wireshark display window. There should be traffic matching the filter being displayed.

b. On R1, issue the command **show flow monitor CCNP-INBOUND-MONITOR statistics**. Your values for the highlighted statistics will be different.

```
R1# show flow monitor CCNP-INBOUND-MONITOR statistics
  Cache type:                               Normal (Platform cache)
  Cache size:                               200000
  Current entries:                               2
  High Watermark:                               12

  Flows added:                                 103
  Flows aged:                                  101
    - Active timeout      (    30 secs)        38
    - Inactive timeout    (    15 secs)        63
```

c. On R1, issue the command **show flow monitor CCNP-INBOUND-MONITOR cache**. **Note**: Output will vary depending upon how long within the 30-second window traffic has been caching.

```
R1# show flow monitor CCNP-INBOUND-MONITOR cache
  Cache type:                               Normal (Platform cache)
  Cache size:                               200000
  Current entries:                               1
  High Watermark:                               12

  Flows added:                                 112
  Flows aged:                                  111
    - Active timeout      (    30 secs)        43
    - Inactive timeout    (    15 secs)        68

IPV4 SOURCE ADDRESS:        192.168.1.75
IPV4 DESTINATION ADDRESS:   10.0.0.1
TRNS SOURCE PORT:           0
TRNS DESTINATION PORT:      2048
INTERFACE INPUT:            Gi0/0/1
FLOW SAMPLER ID:            0
IP TOS:                     0x00
IP PROTOCOL:                1
ip source as:               0
ip destination as:          0
ipv4 next hop address:      0.0.0.0
ipv4 source mask:           /0
ipv4 destination mask:      /0
tcp flags:                  0x00
interface output:           Null
counter bytes:              12024
```

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```
counter packets:           8
timestamp first:           20:43:34.189
timestamp last:            20:43:41.263
<output omitted>
```

d. Stop all the pings and exit the SSH session.

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